

PARTS LIST

Semiconductors:

IC1-IC2	- PS7141-1A, solid-state MOSFET relay
IC3	- LM2678-ADJ, adjustable switching-voltage regulator
IC4	- LM2576-5, 5V switching-voltage regulator
IC5	- 7809, 9V voltage regulator
IC6	- CA3140, BiMOS operational amplifier
T1	- BU505, npn switching transistor
T2	- BD139, npn transistor
IRF1	- IRFP530 p-channel MOSFET
IRF2	- IRF840 n-channel MOSFET
D1-D2, D4, D6	- BY127, rectifier diodes
D3, D5	- 1N5817, schottky diodes
ZD1	- 4.7V zener diode
ZD2	- 5.8V zener diode
LED1-LED3	- 5mm LED
LED4-LED63	- 5mm white LEDs
BR1	- 1.5A bridge rectifier
Resistors (all 1/4-watt, $\pm 5\%$ carbon):	
R1	- 3.3-kilo-ohm
R2, R9, R15	- 10-kilo-ohm
R3, R4, R10, R12, R14	- 1-kilo-ohm
R5	- 4.7-kilo-ohm
R6	- 270-ohm
R7, R8, R11	- 20-kilo-ohm
R13	- 620-ohm
VR1	- 5-kilo-ohm preset

Capacitors:

C1	- 1000 μ F, 35V electrolytic
C2	- 47 μ F, 35V electrolytic
C3	- 470 μ F, ceramic disk
C4	- 10nF, ceramic
C5	- 220 μ F, 25V electrolytic
C6	- 100 μ F, 25V electrolytic
C7	- 1000 μ F, 16V electrolytic

Miscellaneous:

CON1-CON4	- 2-pin connector terminal
L1	- 68uH inductor
L2	- 100uH inductor
X1	- 230V AC primary to 12V AC, 1.5A secondary transformer
S1	- 17V, 20W solar panel
BATT.1	- On/off switch
	- 2-pin terminal connector
	- 12V, 7Ah battery

The operation of battery charge controller is built around IC6. A threshold of 10.5V is set by adjusting trimmer VR1. When battery voltage is below 10.5V, it indicates that it is under-charged. The power supply for IC6 is 9V provided by IC5. Reference voltage of 4.7V is generated by the combination of resistor R11 and zener diode ZD1.

Part of the battery voltage is applied for comparison at pin 2 of IC6. The threshold level is set with the help of VR1. When battery voltage is lower than 10.5V, output of comparator circuit will be high, which is shown by the on state of green LED



Fig. 5: Actual-size, single-side PCB of the LEDs array

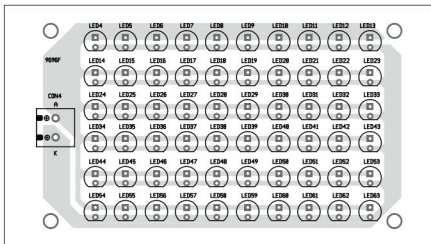


Fig. 6: Component layout of the LEDs array

connected at pin 6 of IC6. When the battery is fully-charged, output of comparator circuit is low. So IRF1 and IRF2 will be switched off, which will prevent the upcoming voltage either from AC panel or solar panel reaching pin 2 of IC3, in turn, preventing overcharging.

When the load is driven by a battery, it becomes an online battery system. When battery voltage is low, AC mains or solar panel comes into picture, as mentioned earlier, and drives the LEDs load and also charges the battery. When the battery is to be charged, the LED load is disconnected by operating switch S1.

The second part of the circuit is the LED array. It is built with 60 LEDs. The lamp is powered by 5V provided by IC4.

Construction and testing

An actual-size, single-side PCB for the solar lamp circuit is shown in Fig. 3 and its component layout in Fig. 4.

Similarly, an actual-size, single-side PCB for the LED array is shown in Fig. 5 and its component layout in Fig. 6.

After assembling the circuit, connect connectors CON3 and CON4 together. Enclose the PCBs in a suitable small box in such a way that the LEDs can light the room well. Ensure proper wiring to avoid any mistake. ●



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